Energy Module: Model Design

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Introduction

Motivation: What GHIM Brings

	Before	GHIM
Economy	More energy detail, less macro feedback	Energy $\leftrightarrow$ production feedback (KLEM)
Climate	Soft coupling or one-way feedback	Two-way coupling with multiple channels
Technology	Mostly fixed cost assumptions	Two-factor learning (RND + LBD) + cross spillovers
Resolution	$\sim$32 regions, 5-year timesteps	100+ countries, annual resolution
Reporting	Post-hoc translation for IPCC	Direct AR6 IAMC-format output

Design Goals

1 Replicates SSP Baselines

Reproduces SSP trajectories before layering policy.

2 Hard-couple Energy and Economy

KLEM production with native energy accounting.

3 Solve Less, Simulate More

Logit-based design for robust convergence.

4 IAMC-Native Reporting

Model structure mirrors IAMC reporting.

5 Transparent & Reproducible

Open-source, every parameter traceable to raw data.

6 User-Configurable

Easy to adjust resolution, nesting, sector structure.

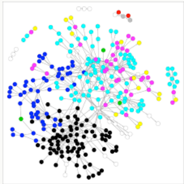
7 Open Economy

Global market clearing for key commodities.

8 Module Linkage

Defined interfaces for other GHIM modules.

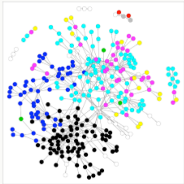
Model Workflow



gcamdata input

SSP · IEA · UN · FAO · CEDS
NREL · GTAP · USGS · EDGAR

Model Workflow



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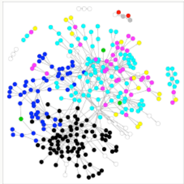
GHIM
Energy Module

GHIM
Other Modules

Model Workflow



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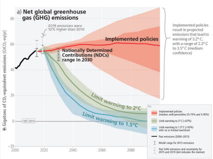


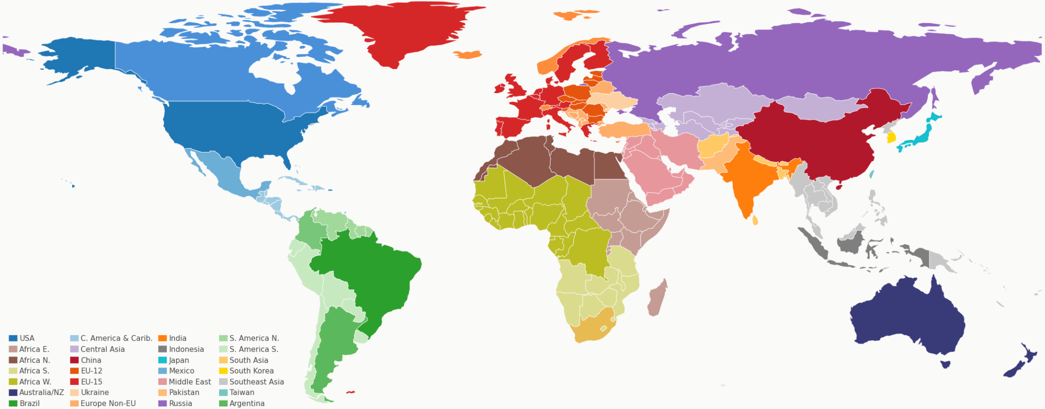
IAMC data template

Framework & Data

General Model Structure by Phase

Component	Phase 1	Phase 2
Time	2000–2150, 5-year steps	Annual resolution
Region	32 GCAM-style regions	100+ countries
Economy	CES-KLE with energy as factor	+ Endogenous energy efficiency via RND
Solution	Recursive-dynamic	Adjustable rolling foresight
Energy System	Minimum nesting for AR6 Tier-1	Expanded nesting for Tier-3
Trade	Global fossil fuel market clearing	Multi-sector global trade
Tech. Choice	Absolute cost logit with SSP-calibrated preference weights	
Tech. Change	Two-factor learning: RND knowledge + cumulative deployment (LBD)	

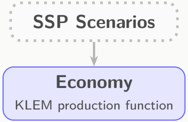
Regional Classification: GCAM 32 Regions



Model Overview

SSP Scenarios

Model Overview



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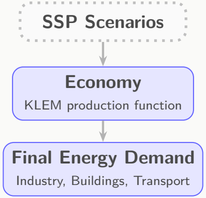

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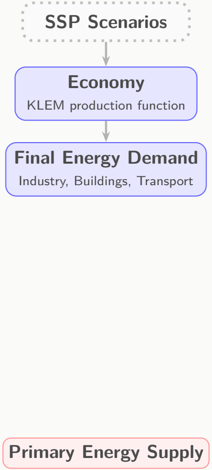

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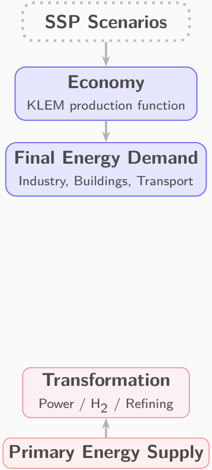

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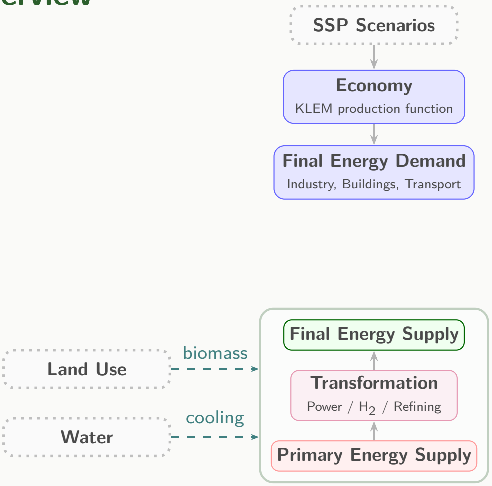

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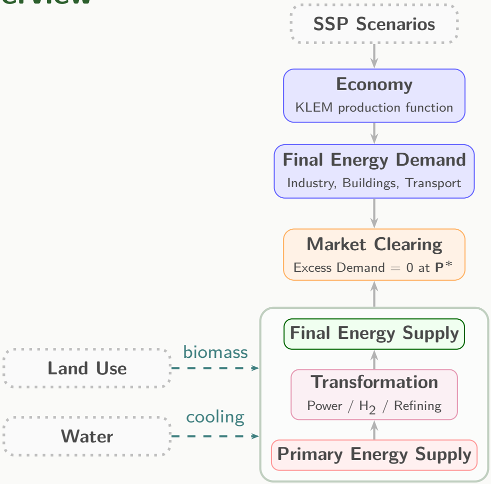

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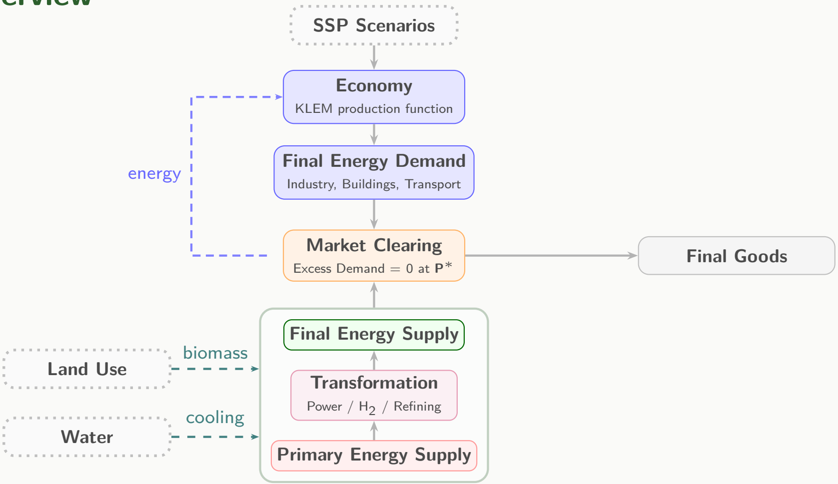

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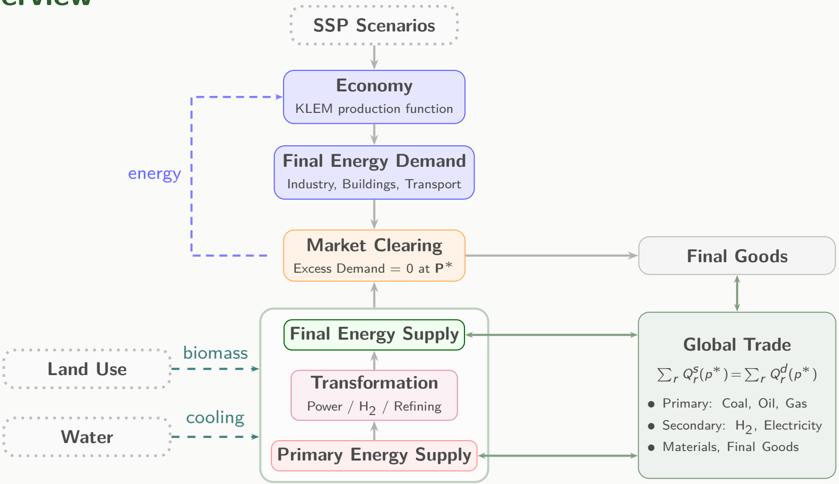

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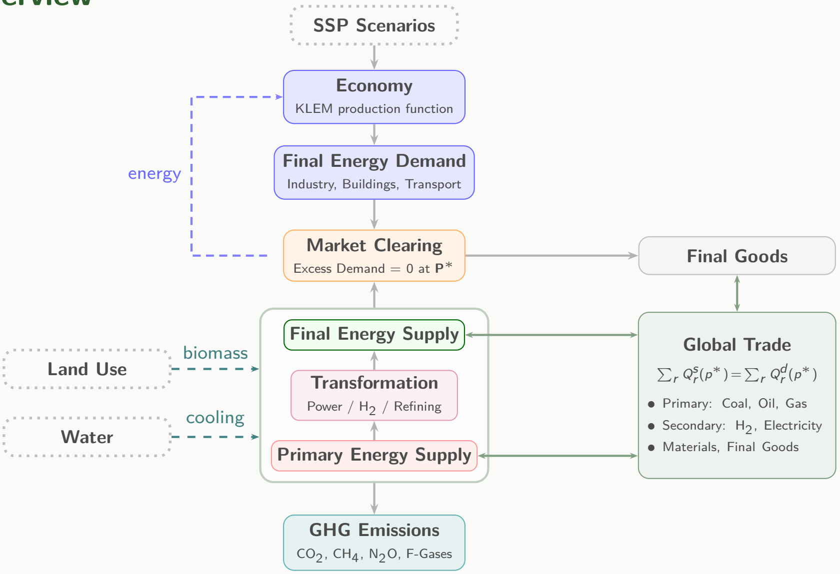

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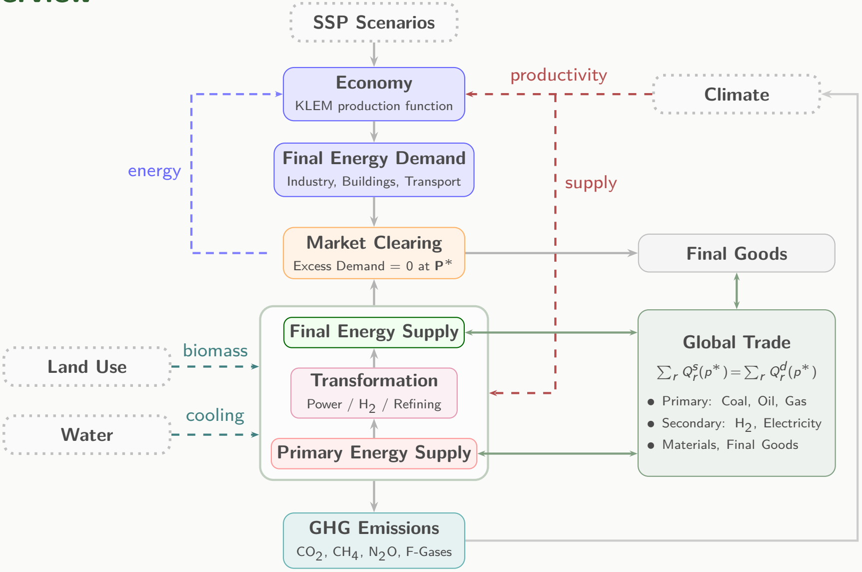
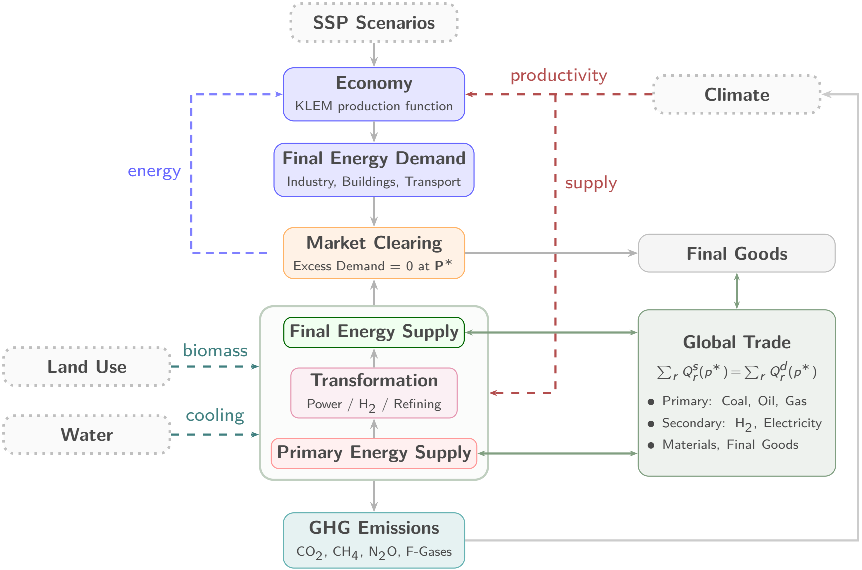
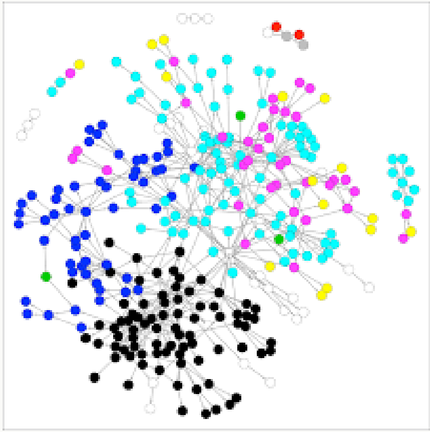



SSP Scenarios
Economy
KLEM production function
productivity
Climate
energy
Final Energy Demand
Industry, Buildings, Transport
supply
Market Clearing
Excess Demand = 0 at $\mathbf{P}^*$
Final Goods
Final Energy Supply
Global Trade
$\sum_r Q_r^s(p^*) = \sum_r Q_r^d(p^*)$
• Primary: Coal, Oil, Gas
• Secondary: H_2, Electricity
• Materials, Final Goods
biomass
Land Use
Transformation
Power / H_2 / Refining
cooling
Water
Primary Energy Supply
GHG Emissions
CO_2, CH_4, N_2O, F-Gases

Input Data: The GCAM Data System



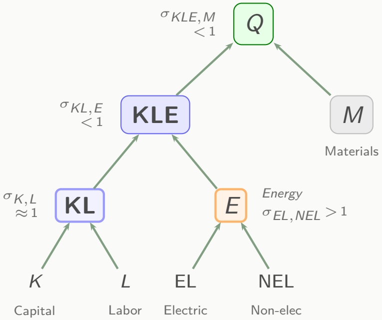
Category	Source	Version
Energy balances	IEA WEB	2023
Scenarios	IIASA SSP	AR6
Emissions	CEDS	2022
Trade	GTAP	v11
Resources	NREL ATB	2024
Transport	UCD	2013
Land / Agriculture	FAO	2024

- IEA: International Energy Agency
- CEDS: Community Emissions Data System
- GTAP: Global Trade Analysis Project
- NREL: National Renewable Energy Lab
- UCD: UC Davis Transportation Database
- FAO: Food & Agriculture Organization

Economy

KLEM Economy — Phase 1 Design

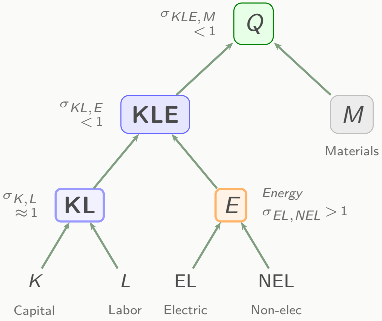
Nesting Structure



- CES: Constant Elasticity of Substitution
- Each nest has its own σ $\longrightarrow$ heterogeneous substitution
- $\sigma < 1$: complements; $\sigma > 1$: substitutes
- $\sigma \approx 1$: unit elastic (Cobb-Douglas)

KLEM Economy — Phase 1 Design

Nesting Structure



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Key Equations

Gross Output: $Q = \mathrm{CES}(KLE,\ M)$

Net output (=GDP): $Y = Q - p_E \cdot E - p_M \cdot M$

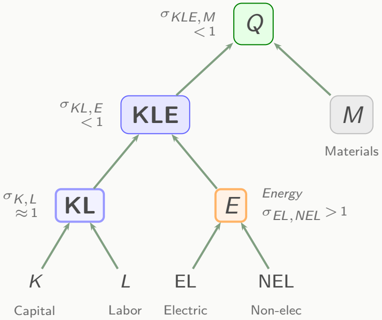
Factor services: $KL = \mathrm{CES}(K,\ L;\ \sigma_{K,L})$

Energy demand: $E = E_0 \cdot \frac{KL}{KL_0} \cdot \left(\frac{p_E}{p_{E,0}}\right)^{-\sigma_{KL,E}}$

Budget: $I_K = s \cdot Y - I_{energy} - \sum_j I_{RD,j}$

KLEM Economy — Phase 1 Design

Nesting Structure



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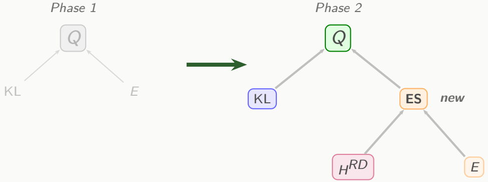
Feedback Mechanisms

- $p_E \uparrow \Rightarrow\ E\downarrow \Rightarrow\ Q\downarrow \Rightarrow\ Y\downarrow$
- $\sigma_{EL,NEL} > 1$: electric $\leftrightarrow$ fossil substitutable
  $\rightarrow$ electrification absorbs fossil price shock
  $\rightarrow$ lower GDP cost of decarbonization

Phase 2 — Energy Services Nest (Aggregate Energy Efficiency Improvement)

Phase 1 → Phase 2



- $I_{RD}^{\mathrm{agg}} = s_{RD} \cdot Y$ (GDP fraction)
- $I_{RD} \to H^{RD}\uparrow$ (knowledge stock)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E
- E: EL + NEL (same as Phase 1)
- $ES = \mathrm{CES}(H^{RD},\ E;\ \sigma_{ES} > 1)$

What is aggregate efficiency?

- Cross-cutting R&D: insulation, smart grid, ...
- Same output, less energy → endogenizes AEEI

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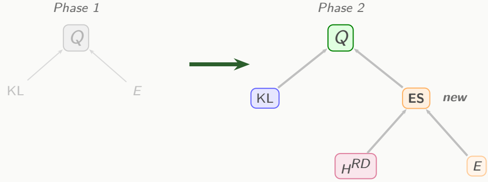
Phase 1 $\rightarrow$ Phase 2



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Energy Service Wraps Physical Energy

Phase 1: $E = ES$ (physical energy = energy service)

- E: physical energy input (EJ $\times$ base price)
- H^{RD}: knowledge stock (R&D-driven efficiency)
- $\sigma_{ES} > 1$: H^{RD} substitutes for E

$\rightarrow$ Same **energy services**, less **physical energy**

$\rightarrow$ Endogenous energy efficiency improvement

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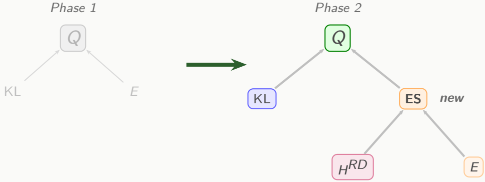

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Unified Budget Constraint

$s \cdot Y = I_K + I_{RD}^{\text{agg}} + \sum_j I_{RD,j}$

- More R&D $\rightarrow$ less K short-term
  but $H^{RD}\uparrow \rightarrow E\downarrow$ long-term

KLEM — Parameter Interpretation

Parameter	Value	Nest	Intuition
σ_{KL}	$\sim$1.0	KL	Unit elastic (Cobb-Douglas)
$\sigma_{KL,E}$	$\sim$0.4	KLE	Complements (<1): energy is essential
$\sigma_{EL,NEL}$	$\sim$2.0	E	Substitutes (>1): electric replaces fossil
$\sigma_{KLE,M}$	$\sim$0.5	Q	Complements (<1): materials essential
TFP	calibrated	KL	Matches SSP GDP trajectory
A_{CES}	calibrated	CES	Ensures $Y =$ GDP at base year
s	0.22	Budget	GDP reinvestment rate
δ	0.05	Capital	Annual depreciation rate
LFP	regional	Labor	ILO 2020 workforce fraction

Calibration — Matching SSP Trajectories

Step 1 — Input Weights α (base year, time-invariant)

Base-year spending fractions on K, L, E, M determine CES weights

Step 2 — Scale Factors A (base year, time-invariant)

Adjust each nest so CES output matches observed base-year values

Step 3 — Productivity TFP(t) (solved at every future year)

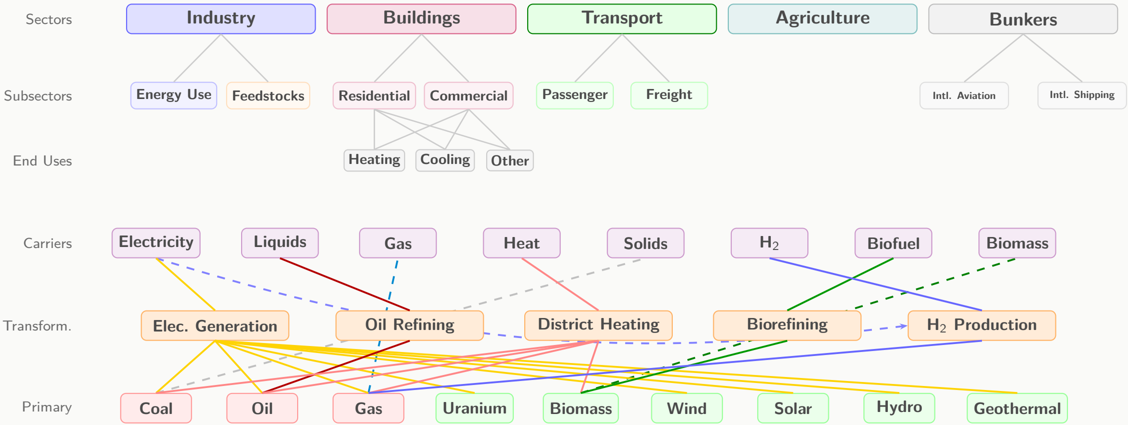
Find TFP(t) so that: $Q(t) - p_E \cdot E - p_M \cdot M = \mathrm{GDP}_{\mathrm{SSP}}(t)$

Result

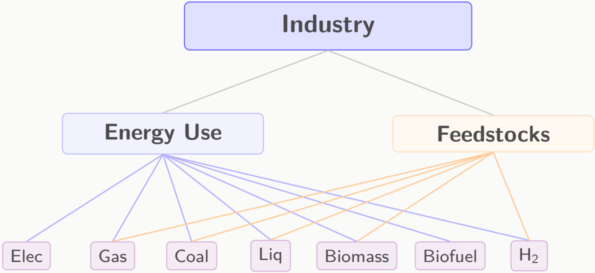
Reference run reproduces SSP GDP within tolerance ($< 0.2\%$ deviation).

Energy System

Energy System Overview (Phase 1)

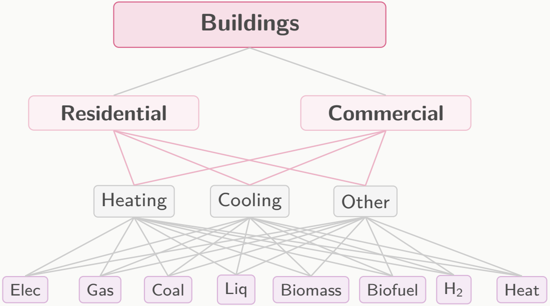


Industry



- **Demand:** $\mathrm{Prod} = A_{\mathrm{prod}} \cdot \mathrm{Prod}_0 \cdot (Y/Y_0)^{\alpha} \cdot (P/P_0)^{\beta}$
- **Energy Use:** $EU = ei_{EU} \cdot \mathrm{Prod}$
- **Feedstocks:** $FS = ei_{FS} \cdot \mathrm{Prod}$
- **Price Index:** $P = ei_{EU} \cdot P_{EU} + ei_{FS} \cdot P_{FS}$

Buildings


Buildings
Residential
Commercial
Heating
Cooling
Other
Elec
Gas
Coal
Liq
Biomass
Biofuel
H_2
Heat

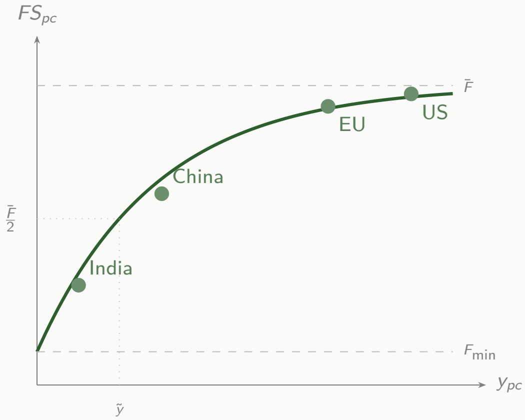
Buildings — Demand (1/2): Floorspace

Step 1: Per-capita floorspace

Saturates with income — rich countries plateau.

$$FS_{pc} = (\bar{F} - F_{\min})\left(1 - e^{\frac{-\ln 2}{\tilde{y}}\, y_{pc}\,(P/P_0)^{\beta}}\right) + F_{\min}$$

- $\bar{F}$: satiation level (m^2/cap)
- $\tilde{y}$: midpoint income (50% of $\bar{F}$)
- $F_{\min}$: subsistence floor
- P/P_0: housing price index (relative to base year)
- β: price elasticity of floorspace
- Floor: $FS_{pc} \geq FS_{pc,\mathrm{base}}$ (no shrinkage)



Step 2: Total floorspace

$$FS(t) = FS_{pc}(t) \times \mathrm{Pop}(t)$$

Buildings — Demand (2/2): Service Intensity & Energy

Step 3: Service intensity per m^2

Thermal load $\times$ service density per unit floor.

$$I_{svc}(t) = \underbrace{DD(t) \cdot U(t)}_{\text{thermal load}} \times \underbrace{d_{svc}\left(\frac{y_{pc}}{P_{svc}}\right)}_{\text{service density}}$$

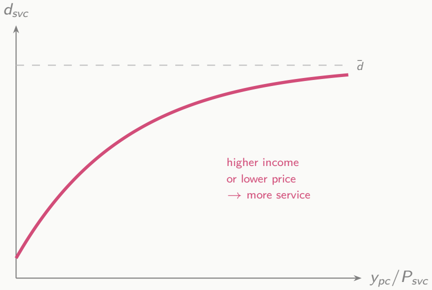
- DD: degree days (HDD for heating, CDD for cooling)
- $U(t)$: shell conductance — improves over time
- d_{svc}: satiation function of *affordability* (y_{pc}/P_{svc})
- Non-thermal (lighting, etc.): no DD term

Step 4: Energy demand by service

$$E_{svc}(t) = FS(t) \cdot I_{svc}(t) \cdot \sum_{c} \frac{s_c}{eff_c^{svc}}$$

- s_c: carrier share from logit
- eff_c^{svc}: service efficiency of carrier c

Service density d_{svc}

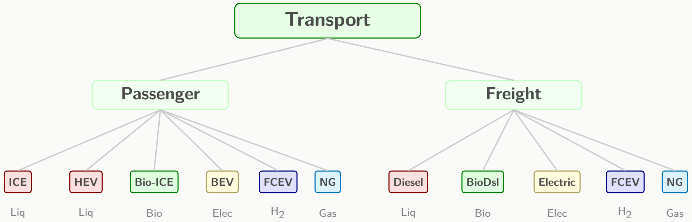


Service price

$P_{svc} = \sum_c s_c \cdot C_c / eff_c^{svc}$ (cost/GJ-service)

- Gas: $eff_{gas}^{svc} \approx 0.9$
- Elec: $eff_{elec}^{svc} \approx 3.0$ (heat pump)

Transport



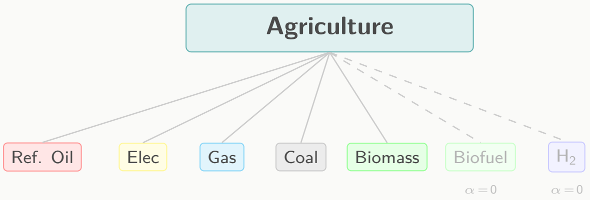
Demand: $SD = sd_{pc,0} \cdot \left(\frac{GDP_{pc}}{GDP_{pc,0}}\right)^{\alpha} \cdot \left(\frac{P}{P_0}\right)^{\beta} \times \text{Pop}$

Energy: $E = \sum_i SD \cdot s_i / eff_i$

LCOT: $\dfrac{capex \cdot FCR}{annual\text{-}SD} + \dfrac{C_{fuel}}{eff} + O\&M + \dfrac{VOT}{speed}$

- *FCR*: annualized capital recovery rate
- *VOT*: value of time (GDP_{pc}/hrs_yr, \$/hr)

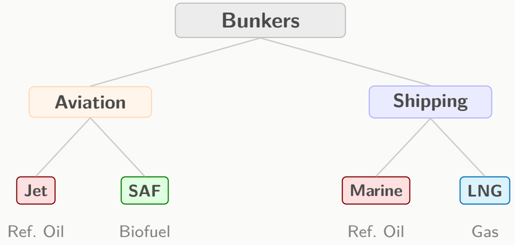
Agriculture



Ag. Output: $Q_{ag} = Q_0 \cdot \left(\frac{GDP}{GDP_0}\right)^{\alpha_{ag}} \cdot \left(\frac{P_{ag}}{P_0}\right)^{\gamma_{ag}}$

Energy: $E_{ag} = ei_{ag} \times Q_{ag}$

Bunkers (International Aviation & Shipping)

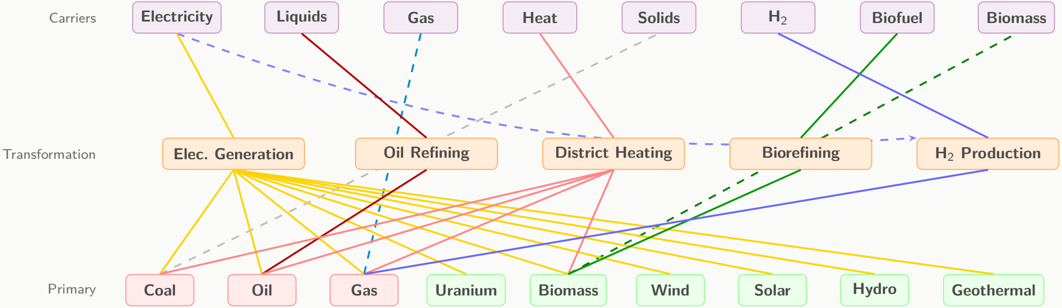


Aviation: $E_{avi} = E_0 \cdot (GDP_{world}/GDP_0)^{\alpha_{avi}}$

Shipping: $E_{ship} = E_0 \cdot (GDP_{world}/GDP_0)^{\alpha_{ship}}$

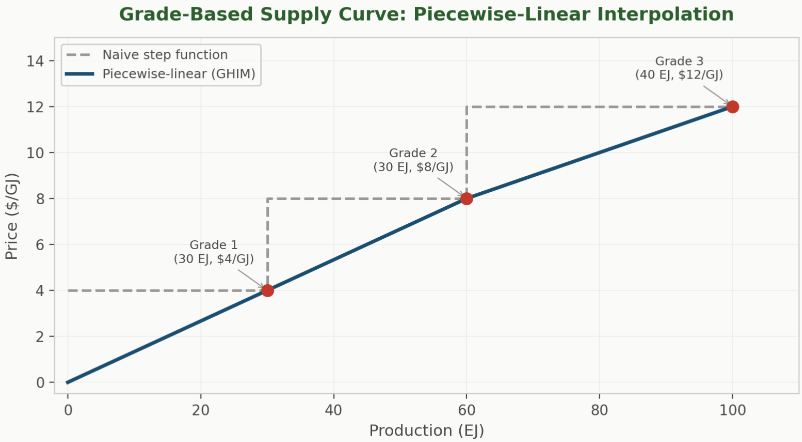
LCOT: $\frac{capex \cdot FCR}{annual\text{-}SD} + \frac{C_{fuel}}{eff} + O\&M$

Energy Supply Chain



Primary Energy Supply — Fossil Fuels

Grade-based supply curves for coal, oil, gas per region — higher prices unlock higher-cost grades.



Inter-Regional Trade — Global Market Clearing

Global Market Clearing

Single world price p^* per fuel (coal, oil, gas)

$\sum_r Q_r^s(p^*) = \sum_r Q_r^d(p^*)$ (supply = demand)

Net trade: $X_r = Q_r^s - Q_r^d$

Calibration Rent (region-specific, fixed at base year)

$\mathrm{rent}_r = P_{\mathrm{observed}} - P_{\mathrm{cleared}}$ per region r

Carried forward as price adder into all future clearing

Supply evaluated at extraction cost + rent_r

Technology Choice

Technology Share

$$s_i = \frac{\alpha_i \cdot e^{\beta\, C_i + p_i}}{\sum_j \alpha_j \cdot e^{\beta\, C_j + p_j}}$$

α_i Availability Switch

Gate $\{0, 1\}$: ban / enable technologies via policy

p_i Preference Weight

SSP-calibrated non-cost factor; decays over time

C_i Cost ($/GJ)

LCOE: capex + O&M + fuel

LCOT: LCOE + time value

Two-Factor Learning

Capital cost declines with cumulative deployment and R&D:

$$\text{capex}_j(t) = \text{capex}_{j,0} \times \underbrace{\left(\frac{Q_j^{\text{cum}}}{Q_{j,0}}\right)^{-\alpha}}_{\text{LBD}} \times \underbrace{\left(\frac{H_j^{\text{RD}}}{H_{j,0}}\right)^{-\beta}}_{\text{RND}} \qquad \text{Floor: } 0.2 \times \text{capex}_0$$

- **LBD** (Learning by Doing): cost ↓ with cumulative deployment Q_j^{cum}
- **RND** (Research & Development): cost ↓ with RND knowledge stock H_j^{RD}

Technology	**LBD Rate**	**RND Channel**
Solar	20%	High
Wind	12%	Medium
Nuclear	3%	Low
Electrolysis	15%	High
Batteries (EV)	10%	High
Heat pumps	10%	Medium
CCS	5%	Medium
H_2 DRI	8%	Medium

Vintage Retirement

Capacity at vintage v survives only if the plant is both young enough **AND** profitable enough.

$$Q_{\mathrm{eff}}(t) = \sum_v C_{i,v} \cdot S_i(t-v) \cdot P(\pi_{i,v})$$

Age-Based Survival $S(\mathrm{age})$

$$S(a) = \frac{1/(1+e^{k(a-\rho L)})}{S(0)}$$

$\rho = 0.75$ (half-life ratio), $k = 0.1$ (steepness)

Lifetime L differs by technology

Renewables: hard cutoff at L (no S-curve)

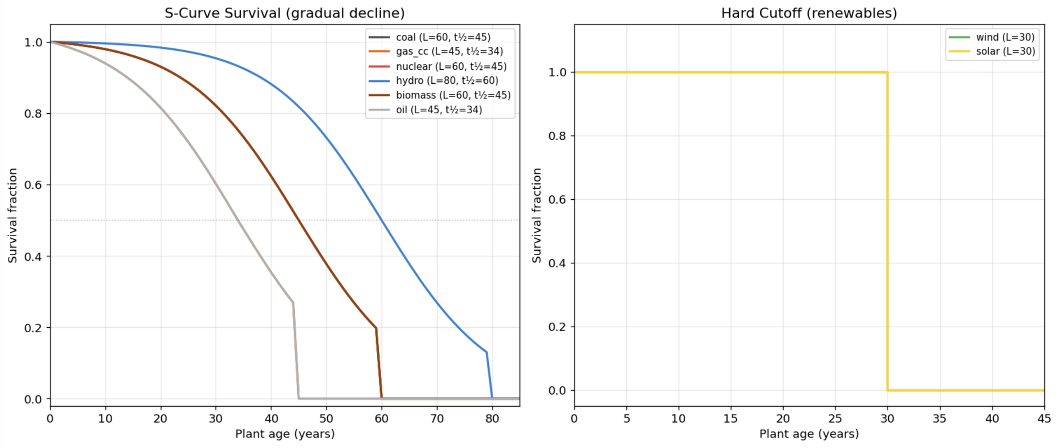
Profit-Based Shutdown $P(\pi)$

$$\pi = (p_{\mathrm{elec}} - c_{\mathrm{var}}) \;/\; |c_{\mathrm{var}}|$$

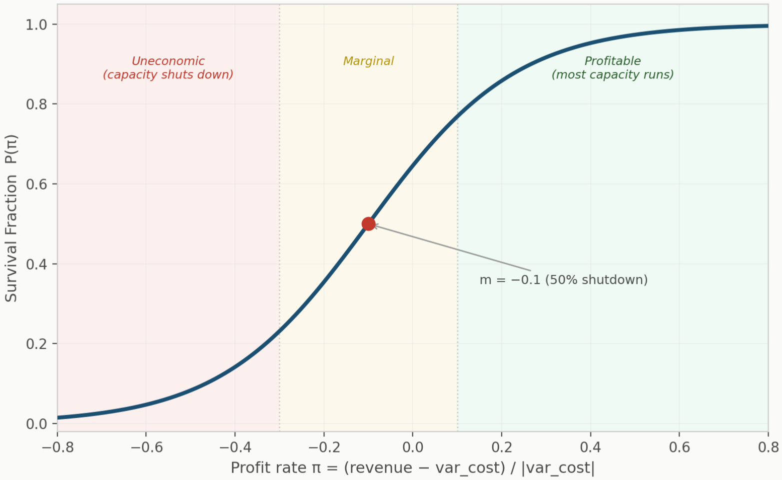
Smooth S-curve in profitability (median $= -0.1$)

Carbon price $\uparrow \rightarrow \pi \downarrow \rightarrow$ coal exit

S-Curve Shutdown



Profit Shutdown



Construction Lead Time

Physical Delay

$$\text{gap} = \text{demand} - \text{surviving} - \text{under-construction}$$

- Under-construction capacity tracked by year
- Nuclear: 2 periods (10yr), Hydro: 1 period (5yr)
- Gap-filling accounts for in-construction capacity

Interest During Construction (IDC)

$$K_{\text{eff}} = K_0 \times (1 + r)^{T_{\text{build}}}$$

- Capital tied up during build earns no return
- Nuclear (10yr, $r = 8\%$): $2.16\times$ capital cost
- Naturally discourages long-build tech in logit

Solver

Solver Design

Approach

1. Recursive-dynamic

Solve one period at a time (2025–2150)

$\mathbf{x}_{n+1} = \alpha\, F(\mathbf{x}_n) + (1-\alpha)\, \mathbf{x}_n$

2. State vector — 133 Dimensions

Regional (×32): $(Y_r,\ I_{E,r},\ p_{\mathrm{elec},r},\ p_{\mathrm{H_2},r})$

Global (×5): $\mathbf{p}^W = (p_{\mathrm{coal}},\ p_{\mathrm{oil}},\ p_{\mathrm{gas}},\ p_{\mathrm{bio}},\ p_{\mathrm{U}})$

3. Implementation

$\alpha = 0.3$ (adaptive)

tol $= 5\times10^{-3}$, max_iter $= 150$

Inline pref recalibration per $F(\mathbf{x})$ call

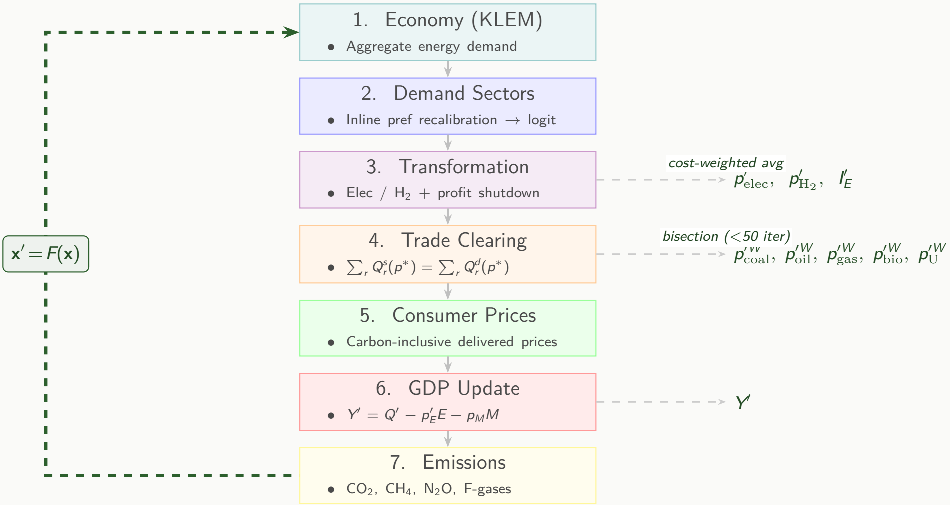
Inter-period demand carry-forward (α_{ind})

Performance

Phase	**Time**
Data loading (Parquet → pandas)	0.9s
TFP calibration	36s
4 passes × 26 periods	
Forward solve	14s
26 periods × ∼0.5s/period	
2 FE recal rounds + inline recal	
per iter: $F(\mathbf{x})$ over 32 regions	
Total	**∼50s**

Single-thread Python. **26/26 periods converge.**

Model Pass $F(\mathbf{x})$

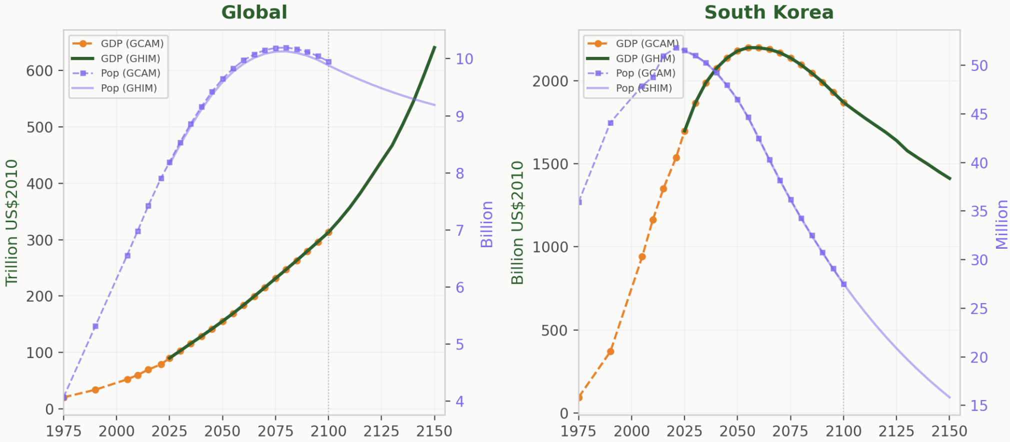


Results & Next Steps

Calibration — GDP & Population

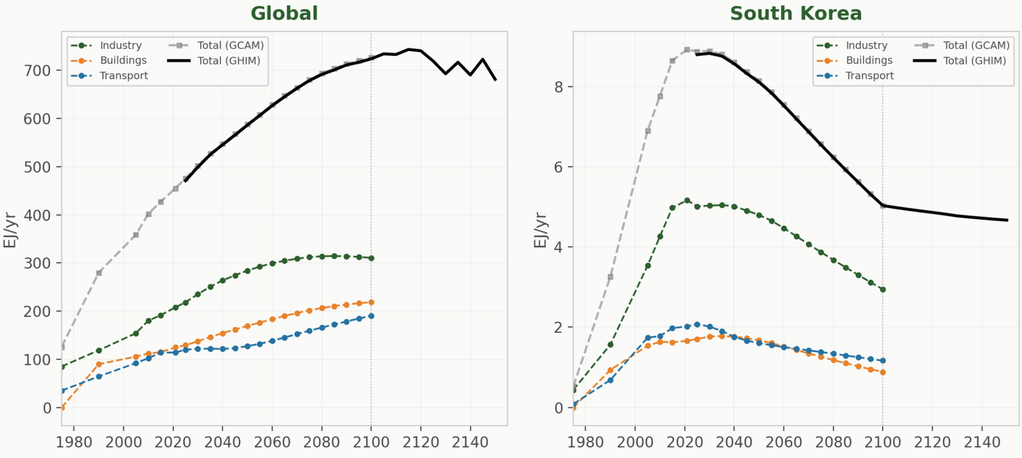
GDP: CC = 1.0000, MAE = 0.3 B$ — TFP reproduces GCAM GDP trajectories

Population: SSP demographic projections applied per R32 region



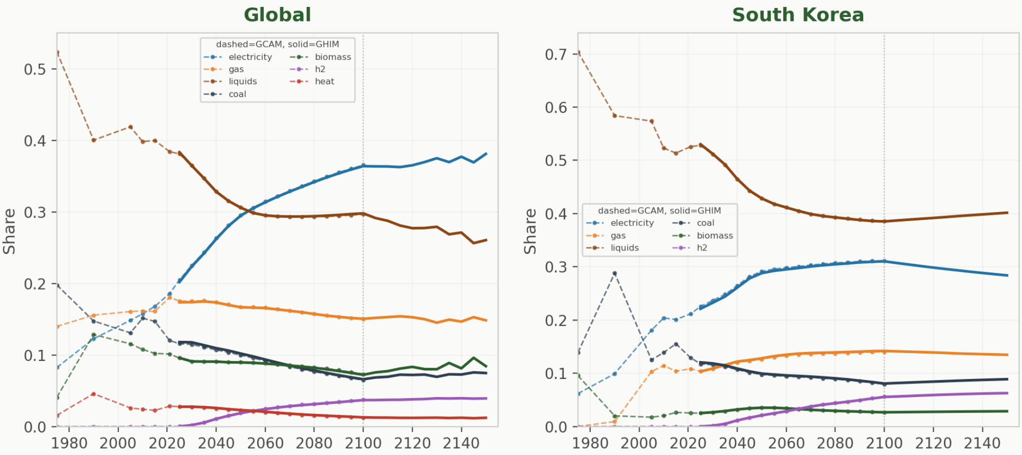
Calibration — Final Energy by Sector

FE: All sectors ±1% (2025–2100). Post-2100: endogenous projection with TFP extrapolation.



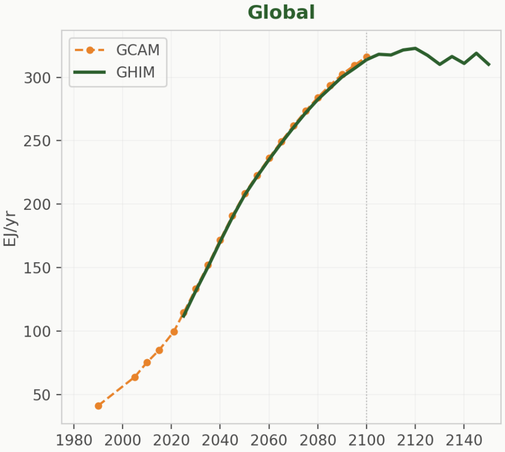
Calibration — Final Energy by Carrier

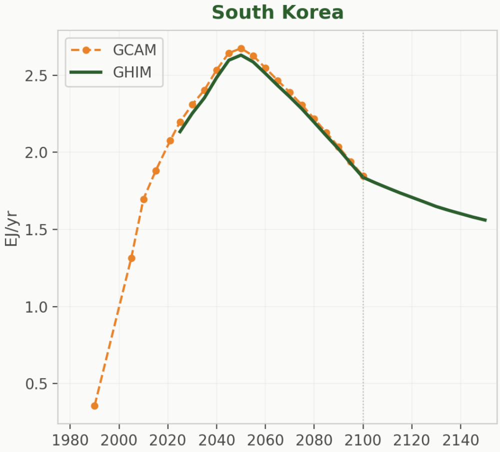
Carrier share trajectories — dashed = GCAM, solid = GHIM. All carriers ±2.7 EJ.



Calibration — Electricity Generation

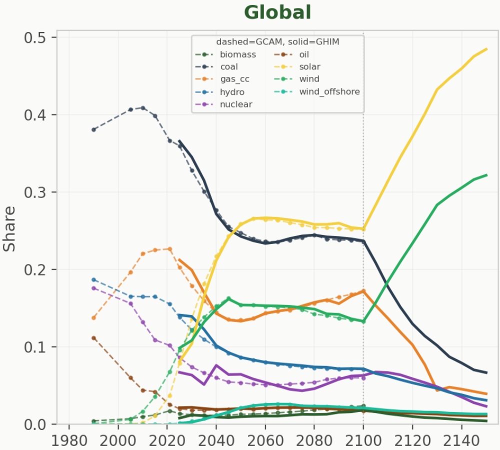
Elec gen: CC = 1.0000, 0.0% deviation all periods. H_2: 0.0%. District heat: 0.0%.

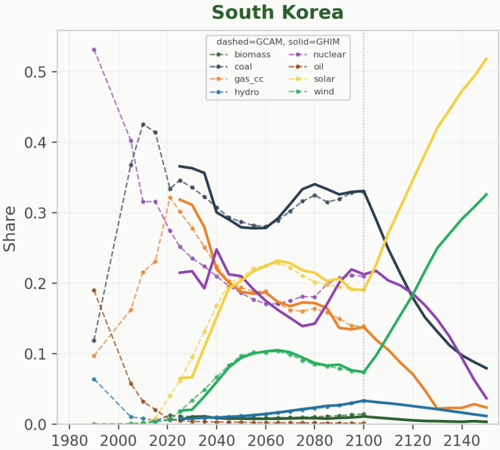




Calibration — Electricity Generation Mix

Generation technology shares — dashed = GCAM, solid = GHIM. All techs ±1 pp.





Next Steps

Component	Phase 1	Phase 2
Time	2000–2150, 5-year steps	Annual resolution
Region	32 GCAM-style regions	100+ countries
Economy	CES-KLE with energy as factor	+ Endogenous energy efficiency via RND
Solution	Recursive-dynamic	Adjustable rolling foresight
Energy System	Minimum nesting for AR6 Tier-1	Expanded nesting for Tier-3
Trade	Global fossil fuel market clearing	Multi-sector global trade
Tech. Choice	Absolute cost logit with SSP-calibrated preference weights	
Tech. Change	Two-factor learning: RND knowledge + cumulative deployment (LBD)	

Thank You

Questions & Discussion

Appendix

Motivation: What GHIM Brings (Detailed)

	Before	GHIM
Economy	· More energy detail, less macro feedback · Coupled models often soft-linked	· Energy $\leftrightarrow$ production feedback (KLEM) · Energy investment crowds out capital · Consistent capital, energy, materials accounting
Climate	· Soft coupling or offline · Mostly through damage function · Often one-sided	· Same-period hard coupling · Climate $\rightarrow$ Economy: multiple channels – TFP damage – Physical: efficiency, resources, demand · Energy $\rightarrow$ Climate via emissions
Technology	· Mostly fixed cost assumptions · Exception: WITCH (optimized RND)	· Two-factor learning (RND + LBD) · Global spillovers across technology families · Energy investment reduces available capital
Foresight	Myopic or perfect foresight	Adjustable rolling horizon
Resolution & Nesting	· ~32 regions, 5-year timesteps · Theoretically flexible · Hard to customize in practice	· 100+ countries, annual resolution · User-configurable sector nesting
Reporting	· Post-hoc translation for IPCC · Custom reporting possible but rarely used	· Direct AR6 IAMC-format output · Flexible reporting with user-defined functions

Formal Calibration (1/4): CES Nesting & Share Parameters

4-Level Nested CES Production Function

Level 1	$\mathrm{KL} = \mathrm{CES}(K,\ L;\ \sigma_{KL}{=}0.8)$	Capital–Labor
Level 2	$E_{\mathrm{val}} = \mathrm{CES}(E_{\mathrm{EL}},\ E_{\mathrm{NEL}};\ \sigma_E{=}2.0)$	Electricity–Non-elec
Level 3	$\mathrm{KLE} = A_{\mathrm{KLE}} \cdot \mathrm{CES}(\mathrm{VA},\ E_{\mathrm{val}};\ \sigma_{KLE}{=}0.4)$	Value-added–Energy
Level 4	$Q = A_{\mathrm{KLEM}} \cdot \mathrm{CES}(\mathrm{KLE},\ M;\ \sigma_{KLEM}{=}0.2)$	Gross output

Share Parameters α_i (base year, time-invariant)

At each nest, α_i is calibrated from base-year cost shares:

$$s_i = \frac{p_i \cdot X_i}{\sum_j p_j \cdot X_j} \qquad \Longrightarrow \qquad \alpha_i = \frac{s_i \cdot p_i^{\sigma-1}}{\sum_j s_j \cdot p_j^{\sigma-1}}$$

KL nest: K, L are factor stocks, so factor prices r, w imputed from income shares ($\alpha_K{=}0.3$).
All other nests: inputs already in value terms (billion \$), so $p_i{\cdot}X_i$ computed directly from data.

Formal Calibration (2/4): Scale Factors A

Scale Factors (base year, time-invariant)

Each nest gets a scale factor so CES output matches observed values:

$$A_{\mathrm{KLE}} = \mathrm{KLE}_0 \;/\; \mathrm{CES}(\mathrm{VA}_0,\, E_{\mathrm{val},0}) \qquad \text{where } \mathrm{KLE}_0 = \mathrm{GDP}_0 + p_E \cdot E_0$$

$$A_{\mathrm{KLEM}} = Q_0 \;/\; \mathrm{CES}(\mathrm{KLE}_0,\, M_0) \qquad \text{where } Q_0 = \frac{\mathrm{GDP}_0 + p_E \cdot E_0}{1 - m}$$

Base-year input computation

$E_{\mathrm{val},0} = p_E \cdot E_0$: energy expenditure ($p_E$ = expenditure-weighted avg of carrier prices, currently global)

$\mathrm{VA}_0 = \mathrm{GDP}_0 = r \cdot K + w \cdot L$: factor income, enforced by TFP calibration

$\mathrm{KLE}_0 = \mathrm{GDP}_0 + E_{\mathrm{val},0}$: value-added **plus** energy cost

$M_0 = m \cdot Q_0, \quad Q_0 = \mathrm{KLE}_0 \,/\, (1 - m) \quad (m = 0.45$, materials share)

Formal Calibration (3/4): Base-Year TFP

Base-Year TFP_0 (iterative calibration)

Find TFP_0 such that $\mathrm{GDP} = Q - p_E \cdot E - m \cdot Q = \mathrm{GDP}_0$:

Initial guess: $\mathrm{TFP} = \mathrm{GDP}_0 \,/\, \mathrm{CES}_{KL}(K_0,\, L_0)$

repeat:

$$\mathrm{VA} = \mathrm{TFP} \cdot \mathrm{CES}_{KL}(K_0,\, L_0)$$
$$E = E_0 \cdot \mathrm{VA} \,/\, \mathrm{VA}_0 \qquad \text{(price ratio} = 1 \text{ at base year)}$$
$$\mathrm{GDP}_{\mathrm{out}} = Q(\mathrm{VA},\, E) - p_E \cdot E - m \cdot Q$$
$$\mathrm{TFP} \leftarrow \mathrm{TFP} \times \mathrm{GDP}_0 \,/\, \mathrm{GDP}_{\mathrm{out}}$$

until $|\mathrm{GDP}_{\mathrm{out}} - \mathrm{GDP}_0| \,/\, \mathrm{GDP}_0 < 10^{-10}$

Notes

Converges in 3–5 iterations (Newton-like ratio correction)

At base year, energy price $= p_{E,0}$, so $E \propto \mathrm{VA}$ (no price effect)

Formal Calibration (4/4): Future TFP Trajectory

TFP(t) via Bisection (every 5-year period)

Given: $K(t)$ from capital accumulation, $L(t)$ from SSP population, $p_E(t)$ from calibration pass.

$$\mathrm{VA}(t) = \mathrm{TFP}(t) \cdot \mathrm{CES}_{KL}\big(K(t),\, L(t)\big)$$

$$E(t) = E_0 \cdot \frac{\mathrm{VA}(t)}{\mathrm{VA}_0} \cdot \left(\frac{p_E(t)}{p_{E,0}}\right)^{-\sigma_{KLE}} \qquad \text{(CES first-order condition)}$$

$$\mathrm{GDP}(t) = Q\big(\mathrm{VA}(t),\, E(t)\big) - p_E(t) \cdot E(t) - m \cdot Q$$

Bisect TFP(t) until $\mathrm{GDP}(t) = \mathrm{GDP}_{\mathrm{SSP}}(t)$ (80 iterations, tol $< 10^{-10}$).

Capital evolution between periods

$$K(t+\Delta) = (1-\delta)^{\Delta}\, K(t) \;+\; s \cdot Y_{\mathrm{SSP}}(t) \cdot \Delta$$

$\delta = 5\%$ (depreciation), $s = 22\%$ (savings rate), $\Delta = 5$ years

$K(t)$ is computed **before** TFP bisection at each period

AR6 Preference Calibration (1/3): Electricity Technology Shares

AR6 SSP baselines specify generation shares per technology per period. Invert the logit to recover preference factors $P_i(t)$ that reproduce these shares.

Relative Preference Logit (MERGE-style)

$$S_i = \frac{e^{\beta \cdot C_i^k + P_i}}{\sum_j e^{\beta \cdot C_j^k + P_j}}$$

S_i: generation share, C_i: LCOE (\$/GJ, incl. capital, O&M, fuel, carbon)

$\beta = -4.0$: cost sensitivity, $k = 0.3$: logit scale

P_i: preference factor (unknown to calibrate)

Logit Inversion (every period, per R32 region)

Reference tech $r =$ largest AR6 share, set $P_r = 0$. For each $i \neq r$:

$$P_i = \frac{\beta}{k} \ln\left(\frac{C_i}{C_r}\right) - \frac{1}{k} \ln\left(\frac{S_i^{\mathrm{AR6}}}{S_r^{\mathrm{AR6}}}\right)$$

Applied to all 17 electricity technologies per R32 region. Roundtrip verified: calibrated preferences reproduce AR6 shares within tolerance.

AR6 Preference Calibration (2/3): Demand Carrier Shares

Calibrate demand-side preferences from AR6 carrier shares (R5 data, applied per R32).

Step A: Extract AR6 Carrier Shares

Transport: direct AR6 variables (3 carriers: liquids, electricity, gas). **Industry & Buildings**: residual method (subtract transport to avoid liquids bias):

$$\text{share}_c^{\mathrm{ind+bld}} = (\mathrm{FE}_c^{\mathrm{total}} - \mathrm{FE}_c^{\mathrm{transport}}) \,/\, (\mathrm{FE}^{\mathrm{total}} - \mathrm{FE}^{\mathrm{transport}})$$

Step B: Within-Carrier Tech Splitting

Multiple techs can share one carrier (e.g., ICE and HEV both use liquids). Split by within-carrier cost competition:

$$S_i^{\mathrm{tech}} = S_c^{\mathrm{carrier}} \times e^{-kC_i} \,/\, \textstyle\sum_{j \in c} e^{-kC_j}$$

Step C: Absolute Preference Logit Inversion

Demand sectors use absolute preference logit (no cost exponent):

$$P_i = (C_r - C_i) - \frac{1}{k} \ln\left(\frac{S_i^{\mathrm{tech}}}{S_r^{\mathrm{tech}}}\right)$$

Per subsector: industry, buildings, transport. Reference $r =$ largest-share tech, $P_r = 0$.

AR6 Preference Calibration (3/3): Final Energy Level Scaling

Shares are correct, but total FE level may not match AR6 (prices change during solving).

Demand Scale Factor $\lambda_s(t)$

Per-sector multiplicative correction to match AR6 total FE:

$$\lambda_s(t) = \mathrm{FE}_s^{\mathrm{AR6}}(t) \;/\; \mathrm{FE}_s^{\mathrm{model}}(t)$$

- $\mathrm{FE}_s^{\mathrm{model}}$: computed at SSP GDP with $\lambda = 1$ (unscaled)
- Applied as: $\mathrm{FE}_s = \lambda_s \times \mathrm{FE}_s^{\mathrm{base}}$
- Shares preserved —only total level changes, not carrier mix

3× Recalibration Loop per Period

- **Round 1**: Compute λ_s from initial prices, run solver
- **Round 2**: Recompute λ_s at converged prices, re-solve
- **Round 3**: Final check, solver re-converges

Each round anchored to SSP GDP. λ_s stabilizes within $\pm 1\%$ after round 2.

R5 to R32 downscaling

AR6 data at R5 (5 regions). Downscale to R32 via GDP share: $\mathrm{FE}_{r32} = \mathrm{FE}_{R5} \times (\mathrm{GDP}_{r32} \,/\, \mathrm{GDP}_{R5})$

Calibration Bootstrap (1/2): Exogenous Pass

Problem: circular dependency

$\mathrm{TFP}(t)$ needs $p_E(t)$, $p_E(t)$ needs model run, model run needs $\mathrm{TFP}(t)$

Pass 1: Exogenous GDP (break the loop)

GDP is forced to SSP; run model to discover equilibrium energy prices.

```
model.gdp_mode = EXOGENOUS                                  // bypass CES for GDP
for t in [2025, 2030, …, 2100]:
solve_equilibrium(model, t)
ref_price[t] = record_energy_prices(t)
```

Now use those prices to calibrate TFP properly.

```
for t in [2025, 2030, …, 2100]:
bisect TFP(t) such that:
```

$Q(t) - p_E(t) \cdot E(t) - p_M \cdot M(t) = \mathrm{GDP}_{\mathrm{SSP}}(t)$

Calibration Bootstrap (2/2): Endogenous Verification

Pass 2+: Endogenous GDP (verify convergence)

CES now determines GDP. Check if it matches SSP; iterate if not.

```
model.gdp_mode = ENDOGENOUS          // CES determines GDP
repeat:
solve_all_periods(model)
max_dev = max_{r,t} |GDP(r,t) - GDP_SSP(r,t)| / GDP_SSP
if max_dev < 2%: break
ref_price = record_energy_prices()
re_calibrate_tfp(ref_price)
```

Result

Without bootstrap (constant-price TFP):	7–14% GDP deviation
With two-pass bootstrap:	$<$ 0.2% GDP deviation

Data Processing

Data Sources & Processing Pipeline

Primary Source: gcamdata (v8.2)

Source	Data	Ver.
IEA	Energy balances	2023
IIASA SSP	Scenarios	AR6
CEDS	Emissions	2022
GTAP	Trade	v11
NREL ATB	Tech costs	2024
UCD	Transport	2013
FAO	Agriculture	2024

450+ R chunks $\rightarrow$ harmonized CSV/XML

Processing Pipeline

1. GCAM BaseX Query

XML query on v8.2 SSP2-Ref database output

2. Parquet Export

`national_accounts`, `final_energy`, `electricity_gen`, `population`

3. CalibrationDataset

R32 $\times$ 22 years (1975–2100)

GDP (MER 2010\$), FE (EJ), shares, subsectors

GCAM 1990\$ $\rightarrow$ 2010\$ ($\times$1.5114 BEA deflator)

4. Model Calibration

TFP iterative 4-pass, inverse logit prefs, FE scaling

Total: $\sim$90s (data 3.5s + TFP calibration 36s + solve 50s)

Current Data Inventory

Category	Source	Format	Resolution	Status
GDP, Capital, TFP	GCAM v8.2	Parquet	R32 × 22yr	**Complete**
Population	GCAM v8.2	Parquet	R32 × 22yr	**Complete**
Final Energy (sector×carrier)	GCAM v8.2	Parquet	R32 × 22yr	**Complete**
Electricity Gen (tech mix)	GCAM v8.2	Parquet	R32 × 21yr	**Complete**
Factor Shares (K, L, E)	GCAM v8.2	Parquet	R32 × 22yr	**Complete**
SSP Scenarios	OECD ENV	CSV.gz	Country × 5yr	**Complete**
Fossil Supply Curves	gcamdata	CSV	R32 × 7 grades	**Complete**
Region Mapping	gcamdata	TSV	249 countries	**Complete**
Technology Costs	NREL ATB	CSV	Per tech	**Defaults**
H_2 Production	GCAM v8.2	—	R32	Not extracted
District Heat	GCAM v8.2	—	R32	Not extracted

Model naming: `GHIM-{source}-{scenario}` e.g. `GHIM-GCAM-v8.2-SSP2-Ref`

Additional Data Needs

Proprietary / Purchase Required

[HIGH] IEA World Energy Balances

Country-level energy flows for direct calibration

[MED] IEA Energy Technology Perspectives

Detailed technology cost projections by region

[MED] GTAP Database (v11+)

Bilateral trade matrices for multi-sector trade

[LOW] BNEF / IRENA Cost Data

Up-to-date renewable energy cost curves

Open Data / Student Tasks

[RA] GCAM BaseX: H_2 & Heat

Extract hydrogen and district heating tech mix

[RA] GCAM BaseX: Refining

Crude-to-product conversion by region

[RA] World Bank PPP Conversion

PPP factors for MER-PPP bridge

[MS/PhD] HDD/CDD Climate Data

Projected heating/cooling degree days

[MS/PhD] Transport Mode Share

Passenger/freight modal split by region

Format convention: Parquet (>1MB) · CSV (mappings) · TSV (region classifications)